YIFEI SUN

Principal Platform Engineer · High-Scale APIs, Cloud Infrastructure & Developer Platforms · 13 Years

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SUMMARY

Principal platform engineer with 13 years building mission-critical, high-volume APIs and cloud-native infrastructure at scale. Expert in Java and AWS ecosystems — designed and shipped public-facing APIs serving millions of daily requests with 99.9% availability and sub-50ms p99 latency. Built API gateways from scratch (custom Nginx/Lua rate limiting, Kong with custom auth plugins), orchestrated microservice migrations on Kubernetes, and established platform standards adopted across 10+ teams. Actively leverages AI tools (Claude, Copilot) to accelerate development workflows — from rapid prototyping and code generation to automated testing and architectural exploration. Combines deep hands-on systems engineering with cross-organizational technical leadership and platform thinking.

LEADERSHIP HIGHLIGHTS

- Led cross-team API redesign serving millions of daily requests: p99 5× reduction, 99.9% availability, Kubernetes-orchestrated microservices on AWS
- Drove platform architecture standards and observability practices across 10+ teams, 3 departments via RFCs and design docs
- Built API gateway infrastructure from scratch: custom Nginx + Lua rate limiting, Kong + Linkerd service mesh, custom Keycloak auth plugins (Python)
- Actively uses AI coding assistants (Claude, Copilot) for rapid prototyping, test generation, architectural brainstorming, and code review acceleration

Technologies: Java (Collections, Streams, Concurrency, I/O), Spring Boot, Python, Node.js · **AWS** (EC2, ECS, EKS, Lambda, SQS, SNS, S3, RDS, DynamoDB), Terraform, Ansible · Kubernetes, Docker, Kafka, Redis, PostgreSQL, Cassandra · REST API, gRPC, GraphQL (Netflix DGS) · Nginx, Kong, Linkerd · Microservices, Event-Driven, CQRS · Datadog, Grafana, Distributed Tracing · AI Tools: Claude, GitHub Copilot · CI/CD

EXPERIENCE

CarGurus — Principal Software Engineer / Technical Lead

Oct 2019 – Present

High-Volume API Platform Modernization — Microservices on AWS

- Led 3 principal/senior engineers and partnered with ~10 teams across 3 departments to redesign the core public-facing API behind CarGurus' most-visited page (~8,000 rpm, millions of daily impressions), migrating from monolith to Kubernetes-orchestrated microservices on AWS.
- Architected a new orchestration service consolidating 12 downstream integrations into 5 well-defined platform services with circuit breaking, fault isolation, and idempotent retry — reducing fan-out complexity and simplifying failure modes.
- Built automated parity-testing infrastructure comparing new vs. legacy responses at 50% production traffic, enabling safe, data-driven rollout decisions with zero customer-facing incidents.

  p99: 250 ms → 50 ms (5×) **Availability: 98% → 99.9%** **Deploy: 1 hr → 3 min** (20×)
- Defined platform-wide architecture standards and observability requirements: structured logging, distributed tracing (Datadog), and SLO-based latency budgets per downstream dependency.

GraphQL Federation — Developer-Facing API Layer

- Designed and integrated a Netflix DGS subgraph into CarGurus' federated GraphQL gateway, exposing vehicle delivery data to frontend, backend, and AI consumers from a single developer-friendly graph.
- Implemented schema federation, batch data loaders (N+1 prevention), and JVM warmup strategies to eliminate JIT compilation latency spikes on cold starts.

Digital Commerce Platform — Event-Driven Architecture on AWS

- Led backend platform design for a new commerce product using microservice and event-driven architecture (SQS/SNS for async messaging, Kafka for event streaming) with Terraform-managed AWS infrastructure.
- Built infrastructure for rapid product experimentation — enabling quick launch, measurement, and safe retirement of product bets without destabilizing core platform services.

Cross-Org Platform Leadership & AI-Augmented Development

- Established reusable patterns for API design, observability, caching, and distributed orchestration adopted across multiple platform teams; authored design docs, RFCs, and architecture decision records.
- Led 10+ training sessions for engineering teams. Mentored senior engineers on system design, API excellence, and path to Principal-level scope.
- Integrated AI coding assistants (Claude, Copilot) into daily workflow for rapid prototyping, test suite generation, and architectural review — measurably accelerating team velocity on complex migration projects.

Applause — Senior Software Engineer / Technical Lead

Oct 2017 – Oct 2019

API Gateway, Authentication & Identity Platform

- Architected and deployed a production API gateway and service mesh (Kong + Linkerd on Kubernetes) with custom token authentication plugins for Keycloak SSO, written in Python. Unified auth across internal systems, enterprise partner platform, and external community.
- Migrated multiple Java and Node.js services behind the new auth gateway with minimal downtime. Performed Glassfish-to-modern-runtime migration with JVM performance tuning.

Event Sourcing & Microservices Architecture

- Drove design and prototyping of microservice architecture based on event sourcing and CQRS; defined foundational patterns for services, events, and read/write models adopted by new platform domains.
- Partnered with product, support, and customer-facing teams to modernize internal workflows and enterprise API integrations.

Toast Inc. — Senior Software Engineer

Sep 2014 – Oct 2017

Core Platform Services & API Infrastructure — Early Engineer

- Among the earliest engineers during hypergrowth; designed and built core platform services end-to-end (Order Service, Labor Service, foundational REST APIs) that became the basis for Toast's public-facing developer platform.
- Built a custom API gateway in Nginx with Lua scripting for rate limiting, request routing, and traffic management — handling all external API traffic before a dedicated platform team existed.
- Set up the initial AWS environment using Ansible — provisioned infrastructure, deployment pipelines, and monitoring. Created the first automated testing framework, enabling safer releases during rapid product expansion.

Third-Party Platform Integrations

- Led API integrations with key partners (e.g., workforce management), designing reliable bidirectional sync with schema evolution. Partnered with PMs, sales, and support to translate customer needs into platform capabilities.

The MathWorks Inc. — Software Engineer II

Jul 2013 – Aug 2014

Developed internal engineering tools (Java backend) for MATLAB/Simulink workflows. Improved backend automation systems for build/test reliability and developer productivity.

EDUCATION

M.S. in Electrical and Computer Engineering — Northeastern University, Boston 2010 – 2013

B.S. in Information Science — Beijing University of Aeronautics and Astronautics 2006 – 2010